

Design & specification overview

Design specification and proposed material schedule for first-sample review.

SOLID MODEL / PRE-SETTING METAL BODY



PROPOSED MATERIAL ROUTE

925 sterling silver, 18K yellow-gold electroplate at a 3.5 µm target, and colourless round brilliant-cut moissanite.

The solid view shows uncut prong stock. Stone sizes, seating and finish require supplier and sample validation.

DESIGN & MATERIAL SCHEDULE

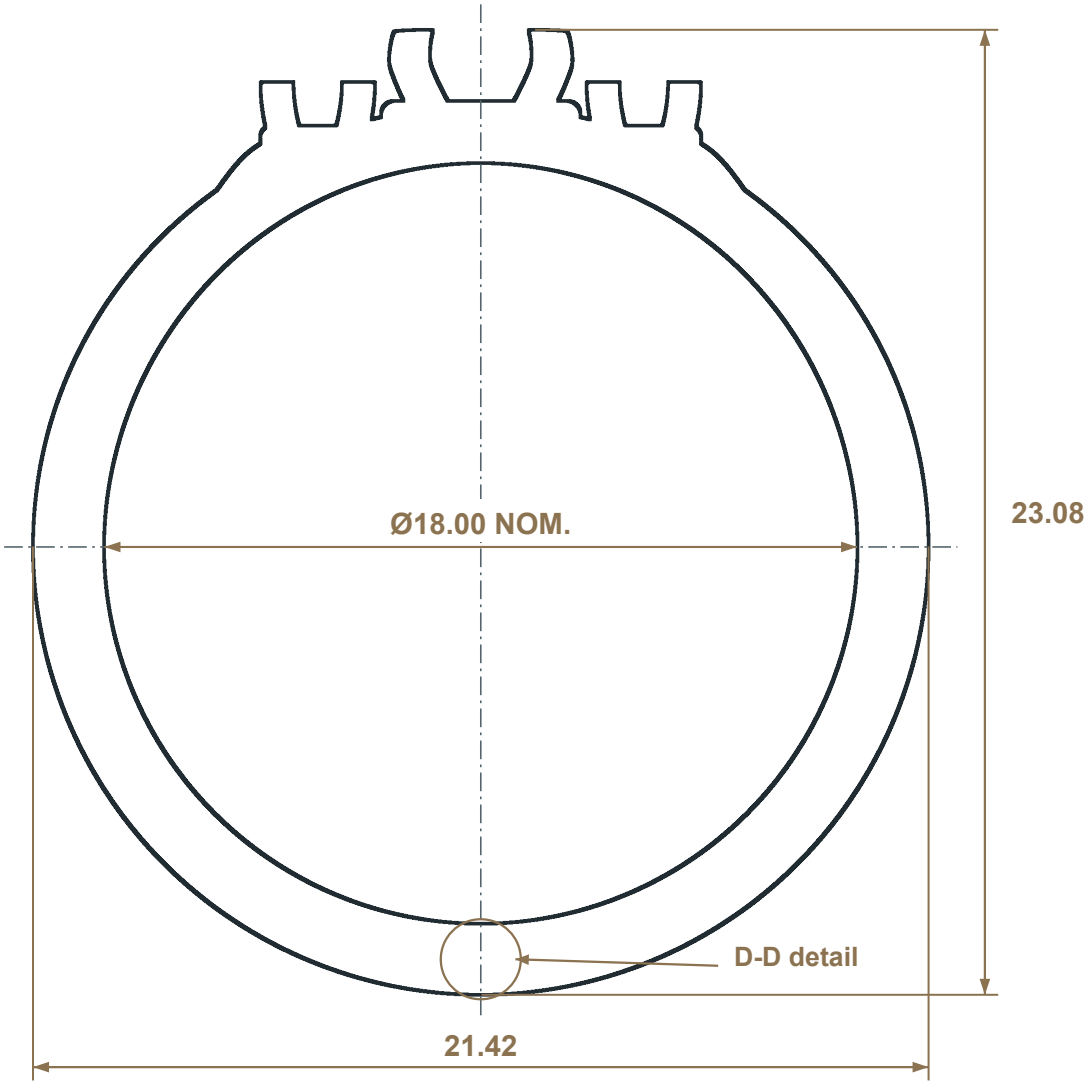
Field	Specification	Basis
Item / type	CF-R01 / bypass three-stone ring	Defined
Overall model body	21.42 W × 23.08 H × 6.00 D	Model measured
Ring bore	18.00 nominal; sampled bore 18.012-18.020	Nominal / sampled
Stone schedule	1 × Ø5.00 + 2 × Ø2.50; 3 total	Nominal model
Metal / plating	925 silver / 18K gold; 3.5 µm target	Proposed
Stone type / cut	Colourless moissanite / round brilliant	Proposed
Findings	12 stock prongs; integrated settings	Model / fit TBC
Finish	High polish, rounded edges and prong tips	Proposed
Target unit cost / MOQ	TBC / supplier quotation required	Open
Release status	Digital study; first sample not yet made	Not released

Assembly dimensions

Orthographic line drawings of the pre-setting metal body, with a measured lower-shank section.

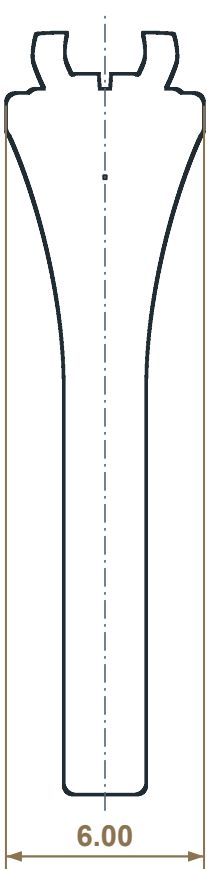
FRONT / X-Z

Projected metal outline and bore



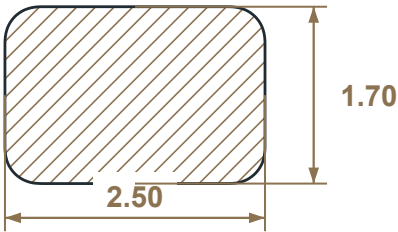
SIDE / Y-Z

Metal envelope; 6.00 mm depth



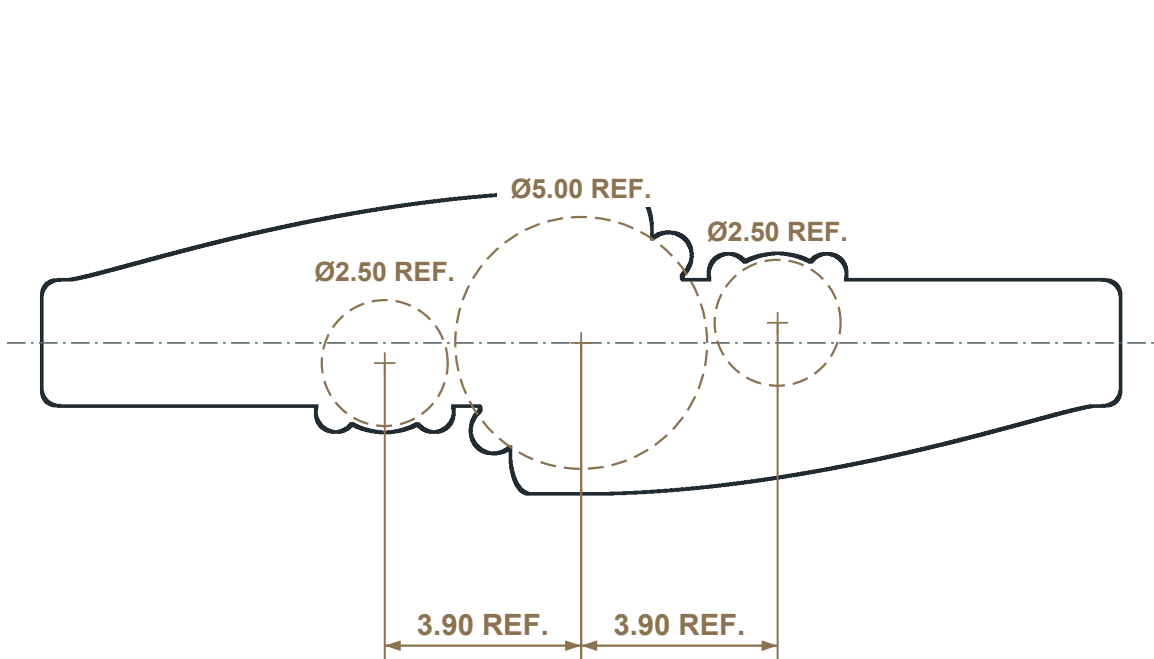
D-D / LOWER-SHANK DETAIL

Local cut at X = 0; dimensions measured in CAD



TOP / X-Y

Metal envelope / nominal stone layout



Dashed circles show nominal stone sizes; dimensioned spacing gives horizontal X offsets. Construction views and gallery details are shown on sheet 03.

LOWER SHANK

2.50 mm axial width × 1.70 mm radial thickness at the indicated section. The enlarged cut shows the actual rounded profile.

A local section describes this location; it does not establish the minimum thickness of every gallery or junction.

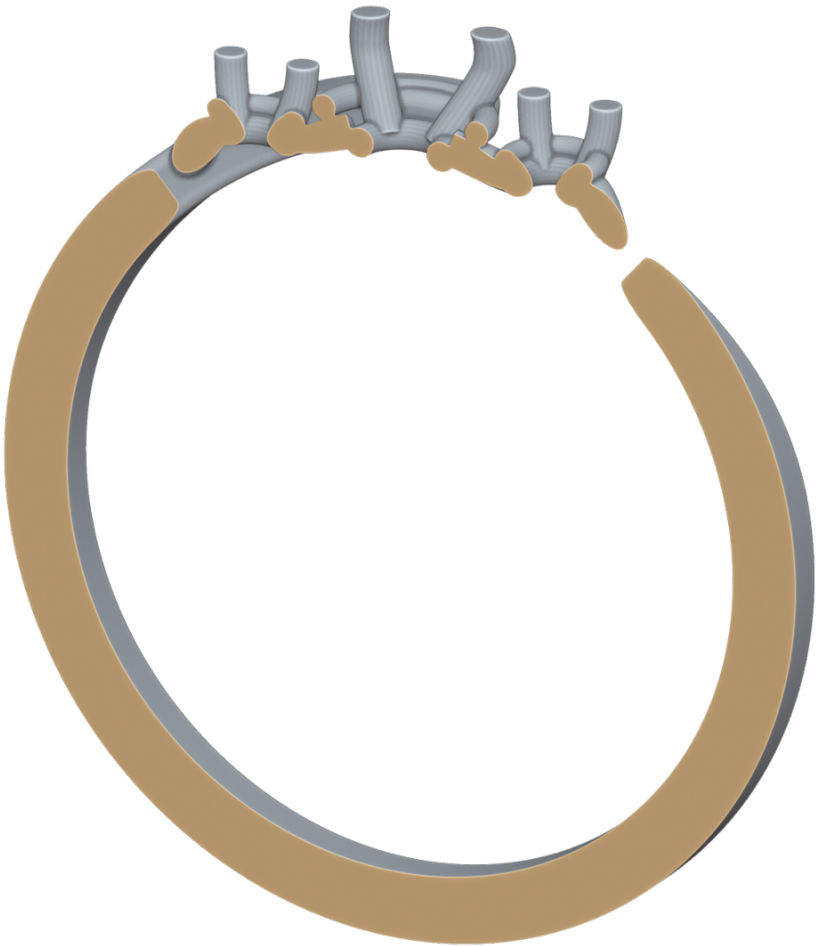
Feature	Value (mm)	Basis
Body envelope	21.42 W × 23.08 H × 6.00 D	Measured pre-setting metal body
Bore / lower shank	Ø18.00 / 2.50 axial × 1.70 radial	Nominal bore / measured local CAD section

Construction & stone setting

A section with 3D context, plus gallery and prong-stock details from the preserved model.

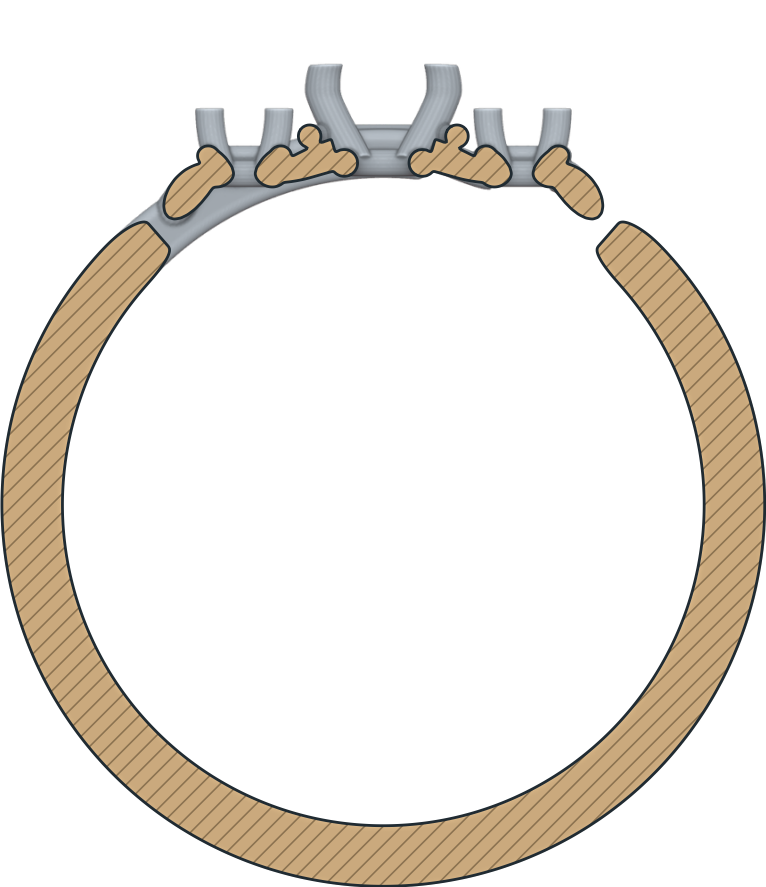
A-A / SLICED 3D MODEL

Y = 0 cut; retain Y ≥ 0

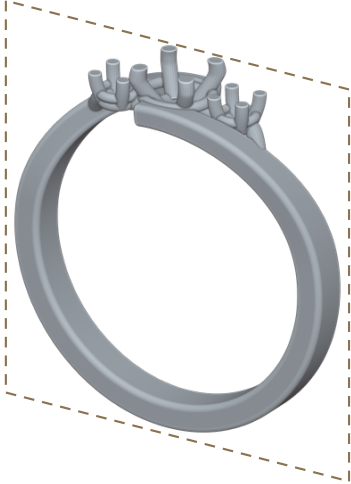


A-A / CUT FACES + RETAINED BODY

Same plane, viewed square-on



SECTION LOCATION / WHOLE MODEL

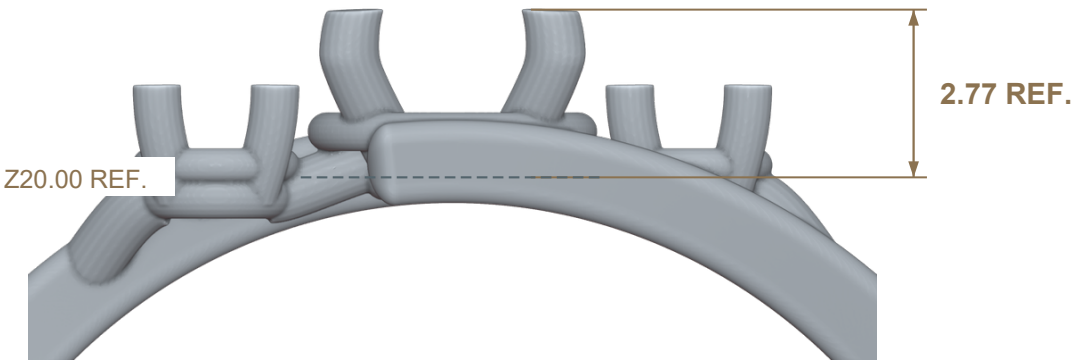


A-A cuts at Y = 0.00 mm, midway through the ring thickness.

The cut crosses gallery and support metal. Several prong axes remain behind the cut.

SETTING / FRONT DETAIL

Uncut stock; final seats to suit purchased stones



Rise is measured above the lower-gallery centreline datum.

Bronze and hatching mark the same five cut regions. Grey metal behind the cut shows their connections. The upper band opening follows the bypass geometry in the removed half.

ID / feature	Nominal / reference geometry (mm)	Construction basis
S01 / centre stone	1 × Ø5.00 × 2.70	Nominal reference profile; purchased-stone fit TBC
S02 / side stones	2 × Ø2.50 × 1.35	Nominal reference profiles; fit TBC
P01 / uncut prongs	4 × Ø0.95 centre; 8 × Ø0.80 sides	Rise: 2.77 / 1.82 REF.; straight-tip Z span ≈0.62 REF.
G01 / upper galleries	OD: Ø4.78 centre / Ø2.76 sides NOM.	Rail stock: Ø0.64 centre / Ø0.56 sides NOM.

Materials, finish & bill of materials

Proposed alloy, stone, coating and surface specifications.

BOM / field	Specification	Acceptance / status
M01 / body	925 sterling silver; one-piece investment casting	Proposed. Supplier alloy certificate and casting suitability required.
M02 / gold layer	18K yellow-gold electroplate; 3.5 µm target	Proposed. Supplier to declare actual stack, process, local minimum and measurement uncertainty.
M03 / underlayer	Composition and thickness TBC	Declare barrier/strike layers; no unapproved substitution. Confirm skin-contact requirements for destination market.
S01 / stones	1 × Ø5.00 round brilliant moissanite	Proposed material; colourless matched lot. Grade, treatment, origin and profile TBC.
S02 / stone matching	2 × Ø2.50 round brilliant moissanite	Proposed material. Measure girdle, total depth, pavilion and diameter range before seats.
F01 / findings	Integrated cast settings	Any solder/weld filler requires process approval.
Finish / polish	High polish on exposed metal; rounded prong tips and contact edges	Maintain gallery access and local section sizes. Finish sample becomes visual master.
Marking / packaging	Maker/material mark and location TBC; individually protected	Verify alloy before marking; include packaging in the quotation.

MATERIAL CHANGE CONTROL

Alternative route: solid 18K yellow gold, without plating; quote separately. Material substitutions require review of casting, setting and finishing requirements.

Manufacturing route & tolerances

Proposed manufacturing stages and first-sample acceptance targets.

Stage / control	Requirement	Verification / responsibility
1 / freeze	Confirm alloy, stone purchase profiles, ring size and source units	Designer + supplier; required before tooling
2 / pattern	Approve a watertight casting master with seat and prong stock matched to purchased stones	Supplier to choose compensated scale, resin, orientation, supports and sprues
3 / cast / prefinish	Investment cast and prefinish inaccessible galleries	Confirm fill, porosity, distortion and remaining section; no blocked cleaning access
4 / stone set	Measure purchased stones; cut/finish bearings; set and close prongs	No glue as a substitute for mechanical retention; verify every seat
5 / finish / plate	Final polish and coating sequence agreed with stone/chemistry compatibility	Supplier to document sequence and protect stones; no unapproved coating stack
T01 / bore	Approved finished bore target ± 0.05 mm	Proposed; confirm the 18.00 mm nominal finished target. Gauge at multiple angles and axial positions.
T02 / overall envelope	Approved finished overall envelope ± 0.10 mm	Proposed; finish allowance/stone height may require revised nominal.
T03 / stone placement	Stone centre positions ± 0.05 mm from approved setting master	Proposed; actual stone grading and setter capability must be checked.
T04 / sections / seats	Minimum finished wall, gallery and prong thickness: TBC	Set by alloy, casting process and seat geometry.
T05 / plating	3.5 μm target; local minimum/upper limit TBC after process trial	Proposed. Calibrated XRF method and measurement sites must be agreed.

Tolerances apply only after a signed sample master exists. Do not shrink or enlarge the production file by an assumed universal percentage. Record tool compensation separately from design dimensions.

Sample review & quality control

First-article inspection criteria and records.

Check	Acceptance criterion	Method / record	Classification
Q01 / identity	Alloy and stone identity match approved BOM	Supplier certificates + suitable material/stone verification; coating can affect surface readings.	Reject substitution
Q02 / size	Bore and envelopes within agreed finished targets	Calibrated gauges; multiple orientations; record measured values and equipment.	Major
Q03 / setting	All 3 stones present; level seating; no movement or chipped girdles	Inspect every stone under magnification; retention test method/load TBC with setter.	Critical if loose/missing
Q04 / prongs	Rounded consistent tips; contour-matched bearings; sufficient metal	Inspect every bearing, stone contact and prong tip under magnification.	Major
Q05 / surface	No burrs, sharp points, visible pits or blocked galleries	Visual master comparison and snag check.	Major / minor by zone
Q06 / coating	Approved thickness, colour and coverage on significant surfaces	Calibrated local XRF check at outer shank, inner bore and accessible setting; sites agreed with supplier.	Major
Q07 / durability	Retention, finish wear and skin-contact performance validated	Supplier to propose quantified wear, sweat and adhesion protocols, limits and reports before testing.	Release gate
Q08 / stacking	No stone-to-metal rubbing; comfortable fit and removal	Wear trial with selected partner rings; photograph clearances and record actual sizes.	Release gate
Q09 / packaging	No transit abrasion; correct item/size/material labelling	Sample pack and transit handling check; stones protected independently.	Major

Defect log fields: sample ID, date, sheet/feature ID, measured result, photograph, severity, cause, correction owner, due date and recheck result. Reject critical faults; batch sampling plan and major/minor acceptance limits remain TBC.

Costing, MOQ & supplier quotation

Quotation requirements and proposed order tiers.

Commercial field	Required value	Supplier response detail
Target unit cost	TBC in NGN per complete ring	Confirm the design budget and supplier quotation.
MOQ / size split	TBC per design / finish / size	Supplier to state minimum order, per-size minimum and mixed-size allowance.
Quote tiers	50 / 100 / 250 rings requested for comparison	Inquiry quantities only. Separate prototype quantity and cost.
Tooling / mold / 3D modelling fees	TBC; currency and ownership stated	Separate one-time charges, tooling life, correction charges and repeat-order rights.
Metal / stones / setting	TBC, each separated	Declare metal weight basis, stone unit costs, setting labour and material wastage.
Polish / coating / QC / packing	TBC, each separated	Declare coating stack and thickness, test/report costs and packing inclusion.
Lead time	TBC: Model approval, tooling, sample, correction, production	State calendar days and the approval/payment trigger for each stage.
Freight / taxes / payment	TBC; Incoterm and destination required	State quote currency, validity, exchange-rate basis, deposit and balance terms.
Defect / rework terms	TBC; written before purchase order	Agree critical defect response, remake/rework costs and claim period.
Model metal volume	265.99 mm³	Supplier to confirm finished metal weight and process yield.

Landed unit cost = metal + stones + setting + finish/coating + QC + packing + allocated tooling + freight/taxes. Record quote date and currency. Compare suppliers only after material, size, stone count, coating and quality requirements match.

Sample approval & release

Open specifications and first-sample approval requirements.

Item	Specification / status	Approval requirement
Drawing revision	CF-R01 / SPEC C	Approve the casting master and finished dimensions before tooling.
Metal body	21.42 W × 23.08 H × 6.00 D mm	Confirm finished wall, gallery and prong thicknesses.
Ring size / units	18.00 mm nominal bore	Confirm nominal ring size and check the finished bore with calibrated gauges.
Materials / stones	925 silver / 18K plate / moissanite proposed	Approve alloy, coating stack, stone grades, profiles and dimensional range.
Sample status	First article TBC	Approve setting, finish, fit and inspection results.
Stack / collection role	Three-stone focal ring; trial beside a plain slender band	Confirm partner-ring clearance and fit in a wear trial.
Size range / resizing	Size range TBC; evaluate lower-shank resizing with supplier	Do not scale stone geometry uniformly; retain purchased stone sizes and reassess settings.
Release approval	Designer: TBC / supplier: TBC / sample ID: TBC	Production release requires signed sample approval.

APPROVAL RECORD

Sample ID	Inspection date	Designer approval	Supplier approval
TBC	TBC	TBC	TBC